

PRATYUSH ACHARYA

AI Researcher | LLM Representations, Interpretability & Alignment

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Research Interests

I study how large language models represent concepts internally, and how to use that understanding to make post-training (SFT, preference optimization, RL) more targeted, producing models that are better aligned and more capable. Topics: representation geometry, mechanistic interpretability (sparse autoencoders, probing, causal interventions, activation steering), interpretability-guided post-training, and training dynamics.

Publications

Concepts Whisper: Spectral Anti-Concentration and the Dual Geometry of Transformer Representations

Pratyush Acharya, Nuraj Rimal, Habish Dhakal

EMNLP 2026 (Main Conference)

[\[arXiv\]](#) [\[Code\]](#)

- Shows that contextual concept directions anti-concentrate in the low-variance tail of the unembedding covariance in all 17 models tested (mean Gini deviation -0.28 ; model-level $p = 3.8 \times 10^{-9}$), replicated on 22 expanded semantic categories and corroborated by SAE features and linear probes.
- Identifies a *dual geometry*: static unembedding contrasts concentrate in high-variance directions while activation-space concepts anti-concentrate. Also shows that Whitened Causal Alignment offers no advantage over spectral regularization for cross-lingual concept transport.

Grokking as a Variance-Limited Phase Transition: Spectral Gating and the Epsilon-Stability Threshold

Pratyush Acharya, Habish Dhakal

Preprint, 2026

[\[arXiv:2603.15492\]](#)

- Models grokking as a variance-gated transition in AdamW optimization, proposing spectral gating and three complexity regimes that govern the shift from memorization to generalization.

Assessing the Pedagogical Readiness of Large Language Models as AI Tutors in Low-Resource Contexts: A Case Study of Nepal's K-10 Curriculum

Pratyush Acharya et al.

Preprint, 2026

[\[arXiv:2604.09619\]](#)

- Introduces a curriculum-aligned benchmark and seven-metric evaluation framework for LLM tutoring, identifying failure modes including the "Expert's Curse" and "Contextual Blindspot."

Education

Tribhuvan University

B.Sc. Computer Science and Information Technology, First Division

Kathmandu, Nepal

2024

Research and Industry Experience

SecurityPal AI

Data Scientist

Kathmandu, Nepal

Sep 2024 - Present

- Fine-tuned a domain sentence-transformer on 19,000 labeled question pairs (TripletLoss + learned pairwise classifier), improving semantic-search accuracy by **21%** and cutting query latency by **3x**.
- Built a reusable LLM evaluation harness that runs **244 security questions** against each knowledge library and attributes failures to content, retrieval, generation, or templating.
- Built and evaluated a Graph-R1 knowledge-graph RAG pipeline (BGE, ChromaDB, Gemini); graph traversal produced **63 correct-answer** and **71 no-hallucination** judgments, opening a path beyond single-shot vector search.
- Shipped a Gemini-based extraction pipeline for multi-format questionnaires (**30-50x speedups**, **40-60 analyst-hours saved per week**) and tool-calling agents that automate multi-step operational workflows.

Invisible Technologies

Advanced AI Trainer - Software Development

Kathmandu, Nepal

Apr 2024 - Aug 2024

- Curated supervised fine-tuning data and built validation pipelines to evaluate LLM-generated code against client quality rubrics for production LLM post-training.

Waterflow Technology

Associate AI Engineer

Kathmandu, Nepal

May 2023 - Nov 2023

- Built a RAG pipeline for document QA, benchmarked open-source LLMs against GPT-series models, and built an Airflow/PySpark ETL pipeline for daily multi-source ingestion.

Teaching and Mentoring

Deerwalk Training Center <i>Trainer</i>	Kathmandu, Nepal <i>Mar 2023 - Present</i>
• Teach data science and machine learning through hands-on projects; designed a beginner-friendly curriculum preparing learners for advanced AI study.	
Deerwalk Mentorship Program <i>Mentor</i>	Kathmandu, Nepal <i>Mar 2023 - Apr 2025</i>
• Mentored participants in object-oriented programming and industry-standard design patterns.	

Technical Skills

Research Methods	Mechanistic interpretability, sparse autoencoders, linear probing, causal intervention, activation steering, spectral analysis
ML and LLMs	PyTorch, Hugging Face Transformers, TRL, SentenceTransformers, LoRA/QLoRA, scikit-learn, DSPy, LangChain
LLM Systems	Evaluation harnesses, RAG and Graph RAG, tool-calling agents, prompt optimization
Languages and Infra	Python, SQL, BigQuery, PostgreSQL, PySpark, Airflow, Docker, Git, Linux